

ALL STABLE MARRIAGES

GUSFIELD'S ALGO A

SAATHWIK AITHAL

24CS01067

I.	Introduction
II.	Stable Matching
III.	Gale-Shapley Algo
IV.	Dry run
V.	Motivation
VI.	Pre-requisites
VII.	Algorithm A
VIII.	Example 1
IX.	Example 2
X.	Insights
XI.	Precedence graph
XII.	Closed subsets
XIII.	Enumerating all
XIV.	Conclusion

INTRODUCTION

Preference List :

Each participant ranks the members of the opposite set.

Men : A, B, C

Women : X, Y, Z

A	Y	X	Z
B	X	Y	Z
C	X	Y	Z

Blocking Pair :

Consider a matching $S : M \rightarrow W$,
a pair (m,w) is called a blocking pair if:

- m prefers w over current partner, and
- w prefers m over current partner.

I.	Introduction
II.	Stable Matching
III.	Gale-Shapley Algo
IV.	Dry run
V.	Motivation
VI.	Pre-requisites
VII.	Algorithm A
VIII.	Example 1
IX.	Example 2
X.	Insights
XI.	Precedence graph
XII.	Closed subsets
XIII.	Enumerating all
XIV.	Conclusion

STABLE MATCHING

From the given preference list
Consider matching,

A ----- X
B ----- Y
C ----- Z

Pair (A,Y) is a **BLOCKING PAIR**.

- A prefers Y over X(current) , and
- Y prefers A over B(current).

A	Y	X	Z
B	X	Y	Z
C	X	Y	Z

X	B	A	C
Y	A	B	C
Z	A	B	C

Stable Matching :

A matching with no blocking pairs.

- Stability prevents conflicts between participants.
- Gale Shapley Algorithm is a famous algorithm to compute a Stable Matching.

I.	Introduction
II.	Stable Matching
III.	Gale-Shapley Algo
IV.	Dry run
V.	Motivation
VI.	Pre-requisites
VII.	Algorithm A
VIII.	Example 1
IX.	Example 2
X.	Insights
XI.	Precedence graph
XII.	Closed subsets
XIII.	Enumerating all
XIV.	Conclusion

ALGORITHM

Input : Two sets (equal size n), Preference lists of both the sets.

```
Set all  $m \in M$  and  $w \in W$  to unmatched
```

```
While there exists unmatched  $m$  :
```

```
(let  $w$  be  $m$ 's most preferred woman that he hasn't proposed yet)
```

```
    if  $w$  is unmatched :  
        match ( $m, w$ )
```

```
    else if match ( $m^*, w$ ) is present :
```

```
        if  $w$  prefers  $m$  to  $m^*$  :  
            unmatched( $m^*, w$ )  
            match ( $m, w$ )  
            set  $m^*$  as unmatched
```

```
        else :  
             $w$  rejects  $m$ 
```

```
end loop
```

I.	Introduction
II.	Stable Matching
III.	Gale-Shapley Algo
IV.	Dry run
V.	Motivation
VI.	Pre-requisites
VII.	Algorithm A
VIII.	Example 1
IX.	Example 2
X.	Insights
XI.	Precedence graph
XII.	Closed subsets
XIII.	Enumerating all
XIV.	Conclusion

DRY RUN

A	X	Y	Z	W
B	Y	Z	X	W
C	Z	X	Y	W
D	W	X	Y	Z

W	B	C	A	D
X	C	A	B	D
Y	A	B	C	D
Z	D	A	B	C

Men optimal
Stable Matching

A	X	Y	Z	W
B	Y	Z	X	W
C	Z	X	Y	W
D	W	X	Y	Z

W	B	C	A	D
X	C	A	B	D
Y	A	B	C	D
Z	D	A	B	C

Women optimal
Stable Matching

A	X	Y	Z	W
B	Y	Z	X	W
C	Z	X	Y	W
D	W	X	Y	Z

W	B	C	A	D
X	C	A	B	D
Y	A	B	C	D
Z	D	A	B	C

No Blocking pairs
Hence, also a Stable
matching

I.	Introduction
II.	Stable Matching
III.	Gale-Shapley Algo
IV.	Dry run
V.	Motivation
VI.	Pre-requisites
VII.	Algorithm A
VIII.	Example 1
IX.	Example 2
X.	Insights
XI.	Precedence graph
XII.	Closed subsets
XIII.	Enumerating all
XIV.	Conclusion

MOTIVATION

- Running the Gale-Shapley algorithm with men proposing produces the man-optimal stable matching (M_0).
- Running the Gale-Shapley algorithm with women proposing produces the woman-optimal stable matching (M_r).
- However, there may exist multiple other stable matchings between these two extreme solutions.
- A naive approach is to enumerate all perfect matchings and test each one for the absence of blocking pairs. Since the number of perfect matchings grows exponentially with n . ($n!$)
- To overcome this limitation, Gusfield's Algorithm A modifies the Gale-Shapley procedure to efficiently discover all rotations, which are then used to enumerate all stable matchings.

I.	Introduction
II.	Stable Matching
III.	Gale-Shapley Algo
IV.	Dry run
V.	Motivation
VI.	Pre-requisites
VII.	Algorithm A
VIII.	Example 1
IX.	Example 2
X.	Insights
XI.	Precedence graph
XII.	Closed subsets
XIII.	Enumerating all
XIV.	Conclusion

PRE-REQUISITES

Stable Pair :

- A stable pair is a man-woman pair (m,w) that appears together in at least one stable marriage.
- In other words, there exists some stable matching in which m is matched with w .

Rotation :

- A rotation is a cyclic sequence of stable pairs that can be simultaneously eliminated to transform one stable marriage into another.
- Eliminating a rotation moves each man to the next woman in the cycle while preserving stability.

$$\pi = ((m_1, w_1), (m_2, w_2), \dots, (m_k, w_k))$$

After eliminating the rotation:

$(m_1, w_1), (m_2, w_2), \dots, (m_k, w_k)$ becomes

$(m_1, w_2), (m_2, w_3), \dots, (m_k, w_1)$

I.	Introduction
II.	Stable Matching
III.	Gale-Shapley Algo
IV.	Dry run
V.	Motivation
VI.	Pre-requisites
VII.	Algorithm A
VIII.	Example 1
IX.	Example 2
X.	Insights
XI.	Precedence graph
XII.	Closed subsets
XIII.	Enumerating all
XIV.	Conclusion

PRE-REQUISITES

Break Marriage :

Breakmarriage is a modified version of the Gale-Shapley algorithm used in Algorithm A.

- Select a man m and break his current marriage. Mark m as free.
- The free man proposes to the next woman on his preference list after his former partner.
- If the woman rejects the proposal, he continues proposing to the next woman.
- If the woman accepts, her current partner becomes free.
- Repeat Steps 2-4 until no free man remains or a man gets rejected by all women.
- Produces a new stable matching by updating engagements through successive proposals.
- Serves as the foundation for Algorithm A to discover rotations and enumerate stable matchings.

I.	Introduction
II.	Stable Matching
III.	Gale-Shapley Algo
IV.	Dry run
V.	Motivation
VI.	Pre-requisites
VII.	Algorithm A
VIII.	Example 1
IX.	Example 2
X.	Insights
XI.	Precedence graph
XII.	Closed subsets
XIII.	Enumerating all
XIV.	Conclusion

ALGORITHM A

```
Find M0 = Man-optimal matching
Find Mr = Woman-optimal matching
```

```
Mi = M0
```

```
while (Mi != Mr)
```

```
    Clear all marks
```

```
    m = first man whose partner in Mi != partner in Mr
```

```
    Break marriage of (m, wife)
    mark wife
    freeMan = m
```

```
    while (true)
```

```
        w = next woman on freeMan's list
```

```
        if w rejects
            freeMan proposes to next woman
```

```
        else
```

```
            if w is not marked
                mark w
```

```
            oldMan = current fiancé of w
            w accepts freeMan
```

```
            freeMan = oldMan
            continue
```

```
    else // marked woman got another proposal
```

```
        // STEP 3a
        Find all women marked since w was marked.
        Their husbands in Mi form ONE rotation.
```

```
        // STEP 3b
        Output this rotation.
        Output the next stable marriage.
```

```
        // STEP 3c
        if (w == original broken wife)
            w accepts freeMan
            break // breakmarriage finished
```

```
        // STEP 3d
        else
```

```
            if w prefers freeMan
                oldMan = current fiancé
                w accepts freeMan
                freeMan = oldMan
```

```
            else
                // reject
                Mark w
                // same freeMan continues proposing
```

```
            continue
```

```
        Mi = newly printed stable marriage
```

```
    end while
```

I.	Introduction
II.	Stable Matching
III.	Gale-Shapley Algo
IV.	Dry run
V.	Motivation
VI.	Pre-requisites
VII.	Algorithm A
VIII.	Example 1
IX.	Example 2
X.	Insights
XI.	Precedence graph
XII.	Closed subsets
XIII.	Enumerating all
XIV.	Conclusion

EXAMPLE 1



A	W	Z	X	Y
B	Y	X	Z	W
C	X	Y	Z	W
D	Y	X	Z	W

W	D	A	B	C
X	A	B	D	C
Y	C	B	A	D
Z	B	D	C	A

Men optimal
Stable Matching

Women optimal
Stable Matching

A ----- X
B ----- Z
C ----- Y
D ----- W



A	W	Z	X	Y
B	Y	X	Z	W
C	X	Y	Z	W
D	Y	X	Z	W

W	D	A	B	C
X	A	B	D	C
Y	C	B	A	D
Z	B	D	C	A

Start with A.
Apply beak marriage.
Mark W.

I.	Introduction
II.	Stable Matching
III.	Gale-Shapley Algo
IV.	Dry run
V.	Motivation
VI.	Pre-requisites
VII.	Algorithm A
VIII.	Example 1
IX.	Example 2
X.	Insights
XI.	Precedence graph
XII.	Closed subsets
XIII.	Enumerating all
XIV.	Conclusion

EXAMPLE 1

X
W

A	W	Z	X	Y
B	Y	X	Z	W
C	X	Y	Z	W
D	Y	X	Z	W

W	D	A	B	C
X	A	B	D	C
Y	C	B	A	D
Z	B	D	C	A

Z
X
W

A	W	Z	X	Y
B	Y	X	Z	W
C	X	Y	Z	W
D	Y	X	Z	W

W	D	A	B	C
X	A	B	D	C
Y	C	B	A	D
Z	B	D	C	A

Z accepts.
Mark Z.
D is free .
D proposes W.
W is marked...

Now its time to go through the 3- a,b,c,d.

Rotation $\pi = ((A,W) (B,X) (D,Z))$

Next matching : A-X , B-Z , C-Y , D-W

As last women to get proposal was first free man's partner
3-c is executed and D matched with W and while loop breaks.

I.	Introduction
II.	Stable Matching
III.	Gale-Shapley Algo
IV.	Dry run
V.	Motivation
VI.	Pre-requisites
VII.	Algorithm A
VIII.	Example 1
IX.	Example 2
X.	Insights
XI.	Precedence graph
XII.	Closed subsets
XIII.	Enumerating all
XIV.	Conclusion

EXAMPLE 2

Women opt matching : 3 6 2 8 1 5 7 4

Last matching : 3 6 1 8 7 5 2 4

W1

1	5	7	1	2	6	8	4	3
2	2	3	7	5	4	1	8	6
3	8	5	1	4	6	2	3	7
4	3	2	7	4	1	6	8	5
5	7	2	5	1	3	6	8	4
6	1	6	7	5	8	4	2	3
7	2	5	7	6	3	4	8	1
8	3	8	4	5	7	2	6	1

1	1	5	3	7	6	1	2	8	4
2	2	8	6	3	5	7	2	1	4
3	3	1	5	6	2	4	8	7	3
4	4	8	7	3	2	4	1	5	6
5	5	6	4	7	3	8	1	2	5
6	6	2	8	5	4	6	3	7	1
7	7	7	5	2	1	8	6	4	3
8	8	7	4	1	5	2	3	6	8

Start with M3.
Mark W1.

I.	Introduction
II.	Stable Matching
III.	Gale-Shapley Algo
IV.	Dry run
V.	Motivation
VI.	Pre-requisites
VII.	Algorithm A
VIII.	Example 1
IX.	Example 2
X.	Insights
XI.	Precedence graph
XII.	Closed subsets
XIII.	Enumerating all
XIV.	Conclusion

EXAMPLE 2

Last matching : 3 6 1 8 7 5 2 4

W7

W2

W1

1	5	7	1	2	6	8	4	3
2	2	3	7	5	4	1	8	6
3	8	5	1	4	6	2	3	7
4	3	2	7	4	1	6	8	5
5	7	2	5	1	3	6	8	4
6	1	6	7	5	8	4	2	3
7	2	5	7	6	3	4	8	1
8	3	8	4	5	7	2	6	1

1	5	3	7	6	1	2	8	4
2	8	6	3	5	7	2	1	4
3	1	5	6	2	4	8	7	3
4	8	7	3	2	4	1	5	6
5	6	4	7	3	8	1	2	5
6	2	8	5	4	6	3	7	1
7	7	5	2	1	8	6	4	3
8	7	4	1	5	2	3	6	8

Start with M3.
Mark W1.

W2 accepts M3.
Mark W2.

W7 accepts M7.
Mark W7.

W2 who was marked, got
proposal

Now its time to go through the 3- a,b,c,d.
Pop out W7 and W2 from stack of marked women

Rotation $\pi = ((M7, W2) (M5, W7))$

Next matching : 3 6 1 8 2 5 7 4

As last women to get proposal was NOT first free man's partner
3-d is executed and ask W2 for her decision.

I.	Introduction
II.	Stable Matching
III.	Gale-Shapley Algo
IV.	Dry run
V.	Motivation
VI.	Pre-requisites
VII.	Algorithm A
VIII.	Example 1
IX.	Example 2
X.	Insights
XI.	Precedence graph
XII.	Closed subsets
XIII.	Enumerating all
XIV.	Conclusion

EXAMPLE 2

Last matching : 3 6 1 8 2 5 7 4

1	5	7	1	2	6	8	4	3
2	2	3	7	5	4	1	8	6
3	8	5	1	4	6	2	3	7
4	3	2	7	4	1	6	8	5
5	7	2	5	1	3	6	8	4
6	1	6	7	5	8	4	2	3
7	2	5	7	6	3	4	8	1
8	3	8	4	5	7	2	6	1

1	5	3	7	6	1	2	8	4
2	8	6	3	5	7	2	1	4
3	1	5	6	2	4	8	7	3
4	8	7	3	2	4	1	5	6
5	6	4	7	3	8	1	2	5
6	2	8	5	4	6	3	7	1
7	7	5	2	1	8	6	4	3
8	7	4	1	5	2	3	6	8

W2 rejects.
So, mark W2.

W1 got proposal now.
W1 is marked.

Now its time to go through the 3- a,b,c,d.
Pop out W2 and W1 from stack of marked women

Rotation $\pi = ((M3,W1) (M5,W2))$

Next matching : 3 6 2 8 1 5 7 4

As last women to get proposal was first free man's partner
3-c is executed and while loop breaks.

INSIGHTS OF ALGO

I.	Introduction
II.	Stable Matching
III.	Gale-Shapley Algo
IV.	Dry run
V.	Motivation
VI.	Pre-requisites
VII.	Algorithm A
VIII.	Example 1
IX.	Example 2
X.	Insights
XI.	Precedence graph
XII.	Closed subsets
XIII.	Enumerating all
XIV.	Conclusion

- Algorithm A discovers all rotations between the man-optimal and woman-optimal stable matchings.
- A single breakmarriage execution may discover multiple rotations.
- Rotations are generated in a valid precedence order, enabling the construction of the rotation precedence graph.
- The total time complexity for finding all rotations is $O(n^2)$.
- As the algorithm progresses from the man-optimal matching (M_0) to the woman-optimal matching (M_r), men receive progressively less-preferred partners, while women receive progressively more-preferred partners.

I.	Introduction
II.	Stable Matching
III.	Gale-Shapley Algo
IV.	Dry run
V.	Motivation
VI.	Pre-requisites
VII.	Algorithm A
VIII.	Example 1
IX.	Example 2
X.	Insights
XI.	Precedence graph
XII.	Closed subsets
XIII.	Enumerating all
XIV.	Conclusion

PRECEDENCE GRAPH

In our Graph,

$\pi_i \rightarrow \pi_j$ means

Rotation π_i must be eliminated before rotation π_j .

Apply the following two rules for all men.

Rule 1:

For each man, consider all his stable partners in preference order.

Example:

M1 : W1 $\rightarrow$ W3 $\rightarrow$ W6

π_1 eliminates (M1,W1)

π_4 eliminates (M1,W3)

π_7 eliminates (M1,W6)

Then add edges : $\pi_1 \rightarrow \pi_4 \rightarrow \pi_7$

I.	Introduction
II.	Stable Matching
III.	Gale-Shapley Algo
IV.	Dry run
V.	Motivation
VI.	Pre-requisites
VII.	Algorithm A
VIII.	Example 1
IX.	Example 2
X.	Insights
XI.	Precedence graph
XII.	Closed subsets
XIII.	Enumerating all
XIV.	Conclusion

PRECEDENCE GRAPH

Rule 2:

- Sometimes a man passes over a woman who is not one of his stable partners.

Example:

Preference List: M1

W1 Stable

W2 Non-stable

W3 Stable

Suppose,

- π_2 eliminates the non-stable pair (M1, W2).
- π_4 eliminates the next stable pair (M1, W3).

Add the edge

$\pi_2 \rightarrow \pi_4$

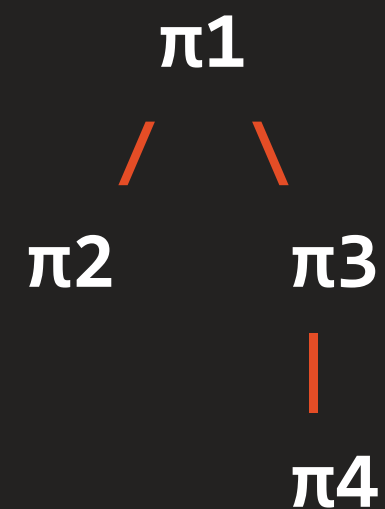
I.	Introduction
II.	Stable Matching
III.	Gale-Shapley Algo
IV.	Dry run
V.	Motivation
VI.	Pre-requisites
VII.	Algorithm A
VIII.	Example 1
IX.	Example 2
X.	Insights
XI.	Precedence graph
XII.	Closed subsets
XIII.	Enumerating all
XIV.	Conclusion

PRECEDENCE GRAPH

- Vertices represent rotations.
- Directed edges represent precedence (dependency) between rotations.
- The graph is a Directed Acyclic Graph (DAG).
- It preserves the precedence information of the complete rotation graph while using only $O(n^2)$ edges.
- The graph is constructed using Rule 1 and Rule 2.
- Every closed subset of this graph corresponds to exactly one stable matching.

CLOSED SUBSETS

The output will be something like:



- Then we write all the closed subsets from the DAG obtained.
- Starting from a null set we write a set of paths from root(s) to the lower nodes but without omitting any predecessor nodes or parent nodes.

Closed Subsets :

- $\emptyset$
- $\{\pi_1\}$
- $\{\pi_1, \pi_2\}$
- $\{\pi_1, \pi_3\}$
- $\{\pi_1, \pi_2, \pi_3\}$
- $\{\pi_1, \pi_3, \pi_4\}$
- $\{\pi_1, \pi_2, \pi_3, \pi_4\}$

I.	Introduction
II.	Stable Matching
III.	Gale-Shapley Algo
IV.	Dry run
V.	Motivation
VI.	Pre-requisites
VII.	Algorithm A
VIII.	Example 1
IX.	Example 2
X.	Insights
XI.	Precedence graph
XII.	Closed subsets
XIII.	Enumerating all
XIV.	Conclusion

ALL STABLE MARRIAGES

- Start from the Man-optimal stable matching (M_o).
- Select a closed subset of rotations.
- Eliminate all rotations in the selected subset while respecting precedence.
- Obtain a unique stable matching.
- Every stable matching corresponds to exactly one closed subset.

Therefore,

Number of Closed Subsets = Number of Stable Matching

I.	Introduction
II.	Stable Matching
III.	Gale-Shapley Algo
IV.	Dry run
V.	Motivation
VI.	Pre-requisites
VII.	Algorithm A
VIII.	Example 1
IX.	Example 2
X.	Insights
XI.	Precedence graph
XII.	Closed subsets
XIII.	Enumerating all
XIV.	Conclusion

CONCLUSION

- Gale-Shapley computes only the man-optimal and woman-optimal stable matchings.
- Algorithm A efficiently discovers all rotations in $O(n^2)$ time.
- Rotations capture the structural differences between stable matchings. The Rotation Precedence Graph (G) represents dependencies among rotations.
- The complete enumeration algorithm runs in $O(n^2 + n \cdot |S|)$ $|S|$ is the number of stable matchings.

REFERENCES

THREE FAST ALGORITHMS FOR FOUR PROBLEMS IN STABLE MARRIAGE.

- DAN GUSFIELD

THANK YOU

ANY QUESTIONS